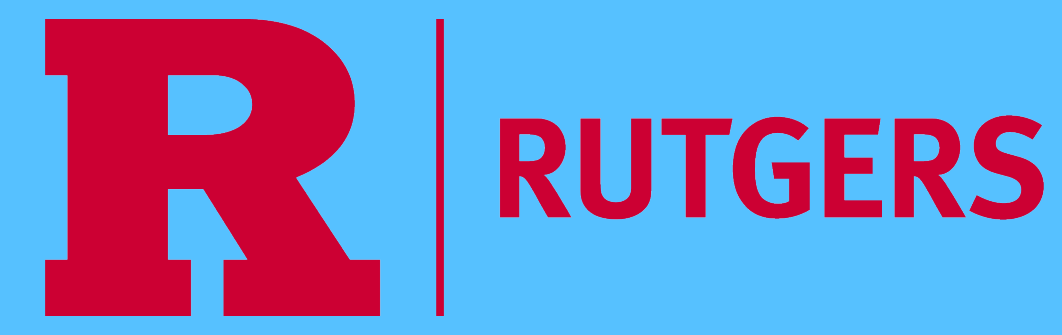


Calculating the Binding Affinities of Lipids to Two pLGICs Using the Density-Threshold Affinity



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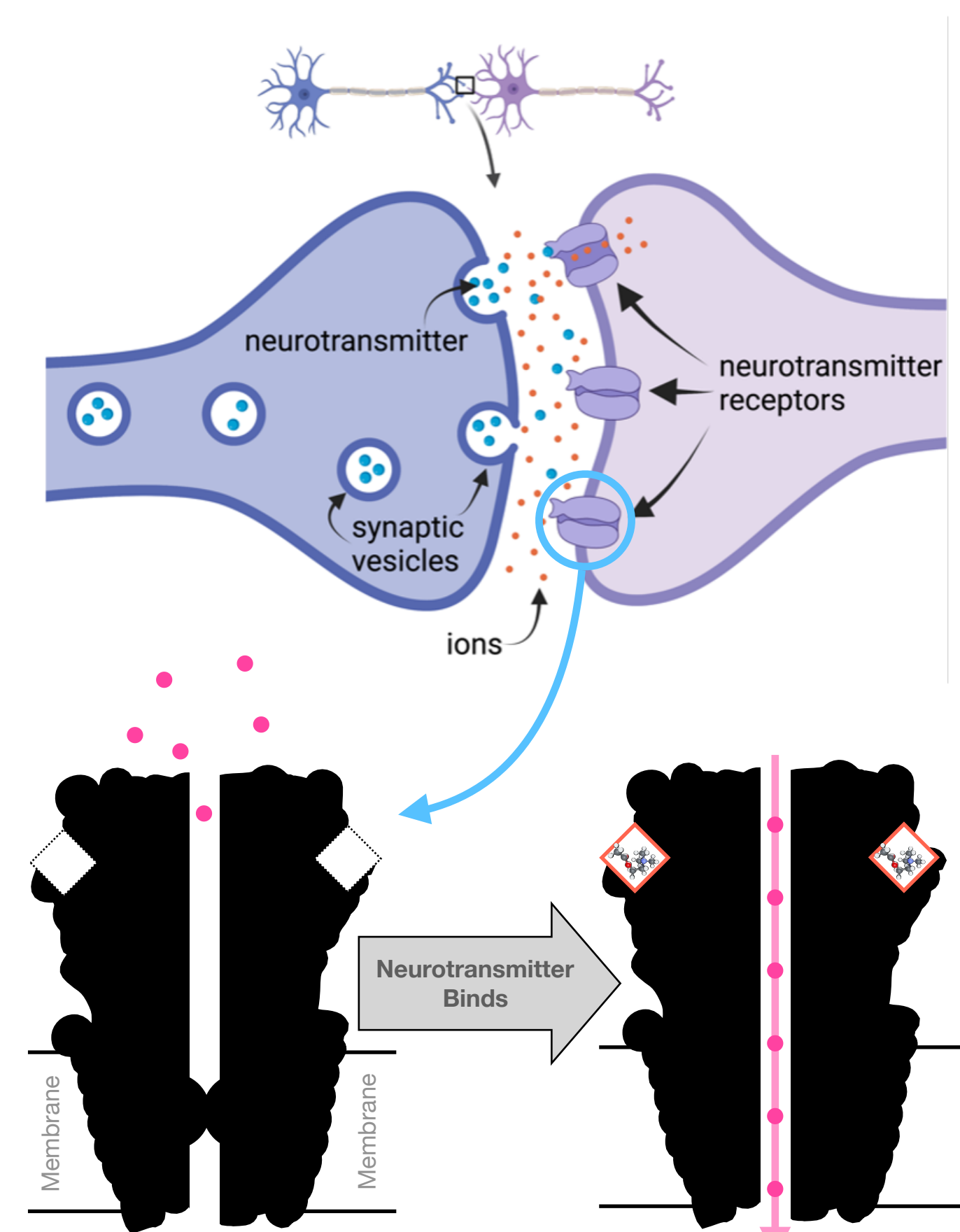
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Abstract

While it has been known for some time that lipids can directly bind to and allosterically modulate membrane proteins, measuring a lipid's binding affinity remains a challenge. Molecular dynamics (MD) simulations represent a promising vehicle for reliably measuring this quantity, but current solutions either utilize complicated biasing procedures or ignore the unreliable kinetics inherent in most coarse-grain (CG) models. We previously introduced the density-threshold affinity (DTA), a technique for measuring a lipid's binding affinity from unbiased CG-MD simulations. In the present work, we use an expanded DTA toolkit to investigate two lipid interactions with pLGICs: arachidonic acid (AA) inhibiting the nicotinic acetylcholine receptor (nAChR), and cardiolipin (CDL) preferentially binding the resting state of a prokaryotic ion channel (ELIC5), potentially stabilizing this state.

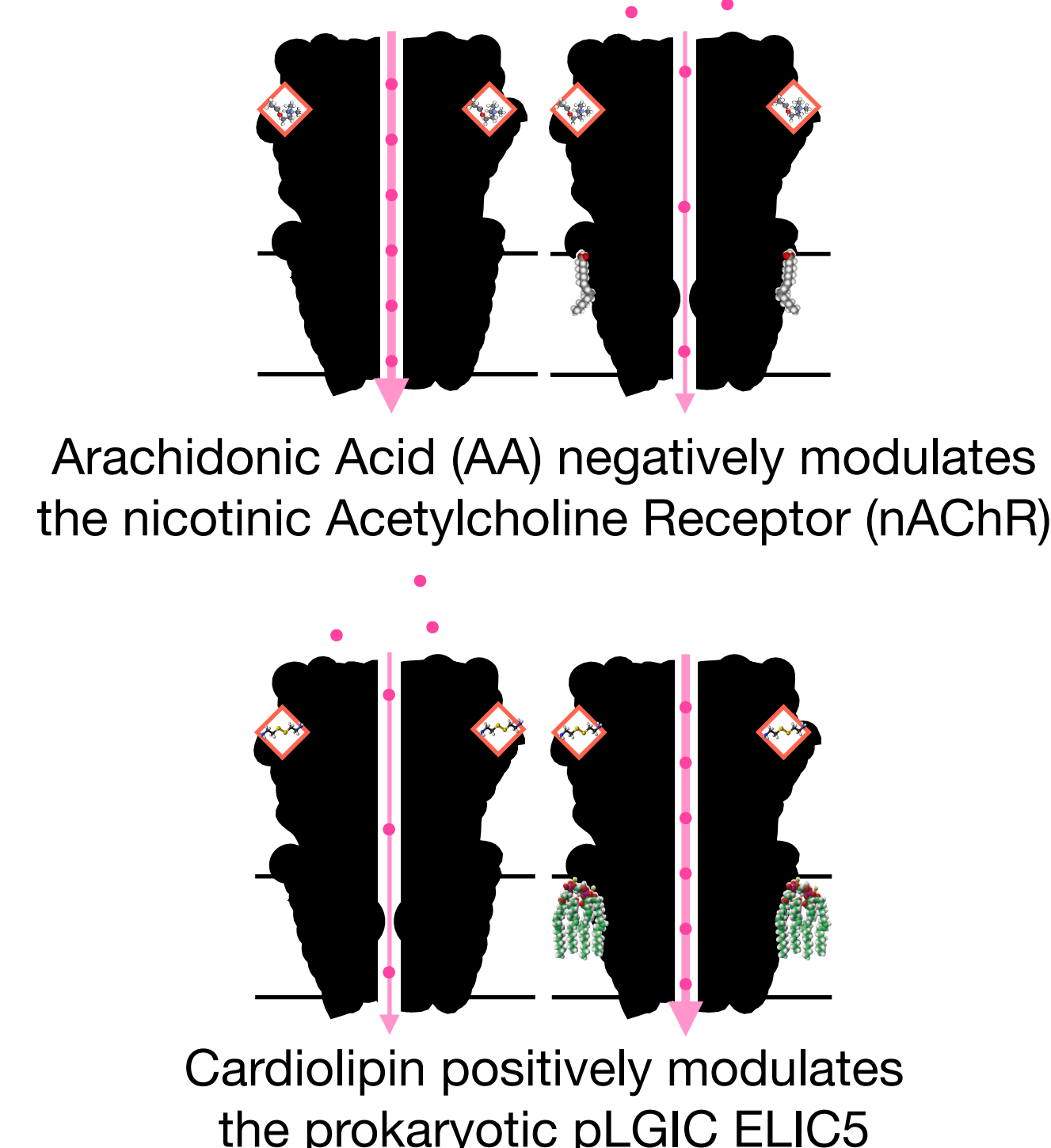
Background



Pentameric Ligand-Gated Ion Channels (pLGICs) relax the stricture in their central pore upon binding of neurotransmitters. This allows for ions to pass through, facilitating signal transmission from one neuron to another.

Motivation

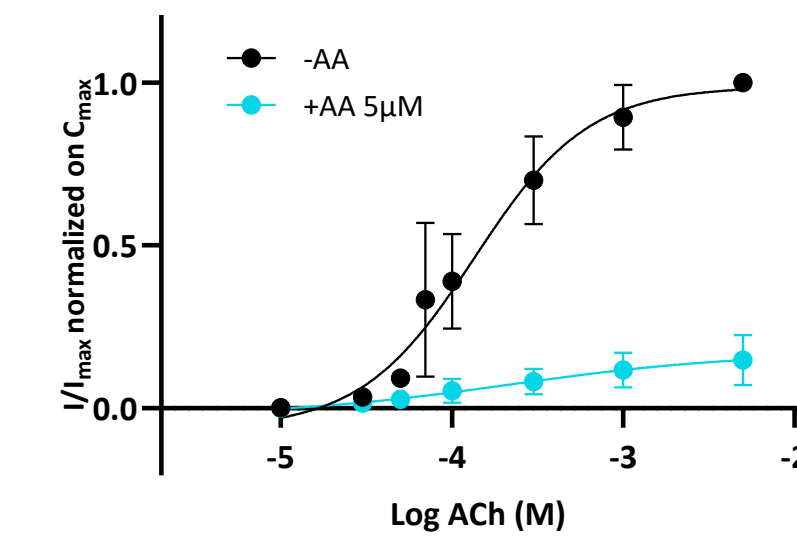
Lipids modulate pLGIC function



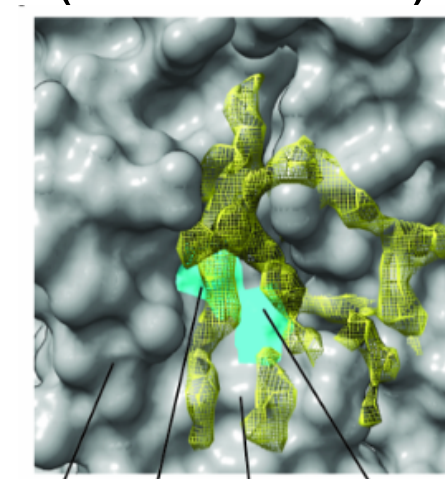
Arachidonic Acid (AA) negatively modulates the nicotinic Acetylcholine Receptor (nAChR)

Cardiolipin positively modulates the prokaryotic pLGIC ELIC5

Wet lab techniques yield partial answers



Electrophysiology can show which lipids are modulators (but not how)

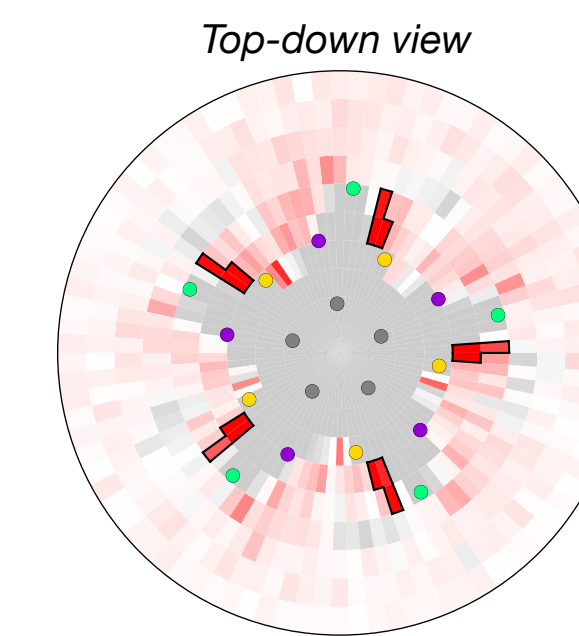


Cryo-EM sometimes shows lipid-like densities (but not which ones)

A physics-based computational approach can tell us:

Which lipids bind? Where? How strongly? Under what conditions?

RESULT 1: Arachidonic Acid (AA) inhibits $\alpha 7$ nicotinic acetylcholine receptor (nAChR) by binding to an intrasubunit site



Density enrichment Polar AA Headgroup Extracellular leaflet

Fig. 1: Density enrichment of AA headgroups measured in a 22-component oocyte-mimetic background. Region of 15-fold enrichment is outlined in black. The Density-Threshold Affinity (ΔG_{bind}) for AA headgroups in that site is -2.3 ± 0.1 kcal/mol.

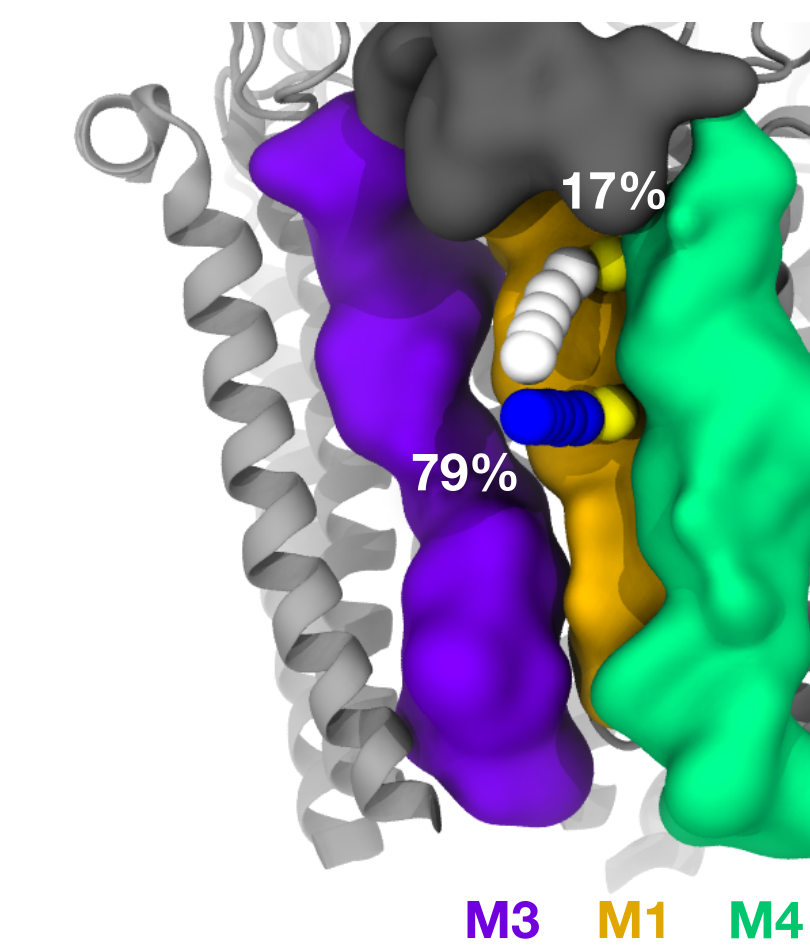


Fig. 3: Most occupied binding modes depict AA head groups embedded in the intrasubunit crevice between the M1 and M4 helices.

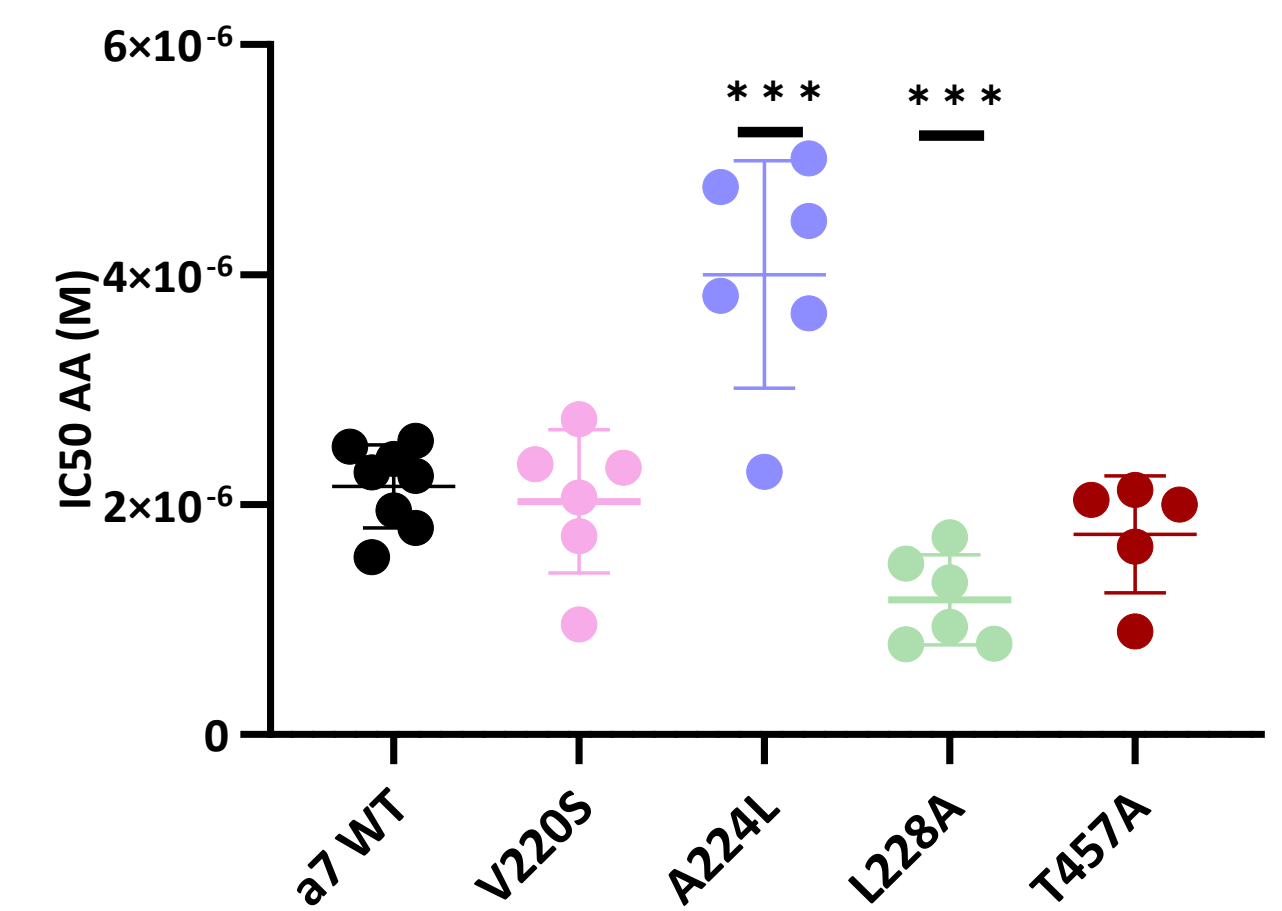


Fig. 5: Mutagenesis results. Two mutants, A224L and L228A, significantly shifted the IC50 of AA, supporting a functional role for those residues (** $p=0.0004$).

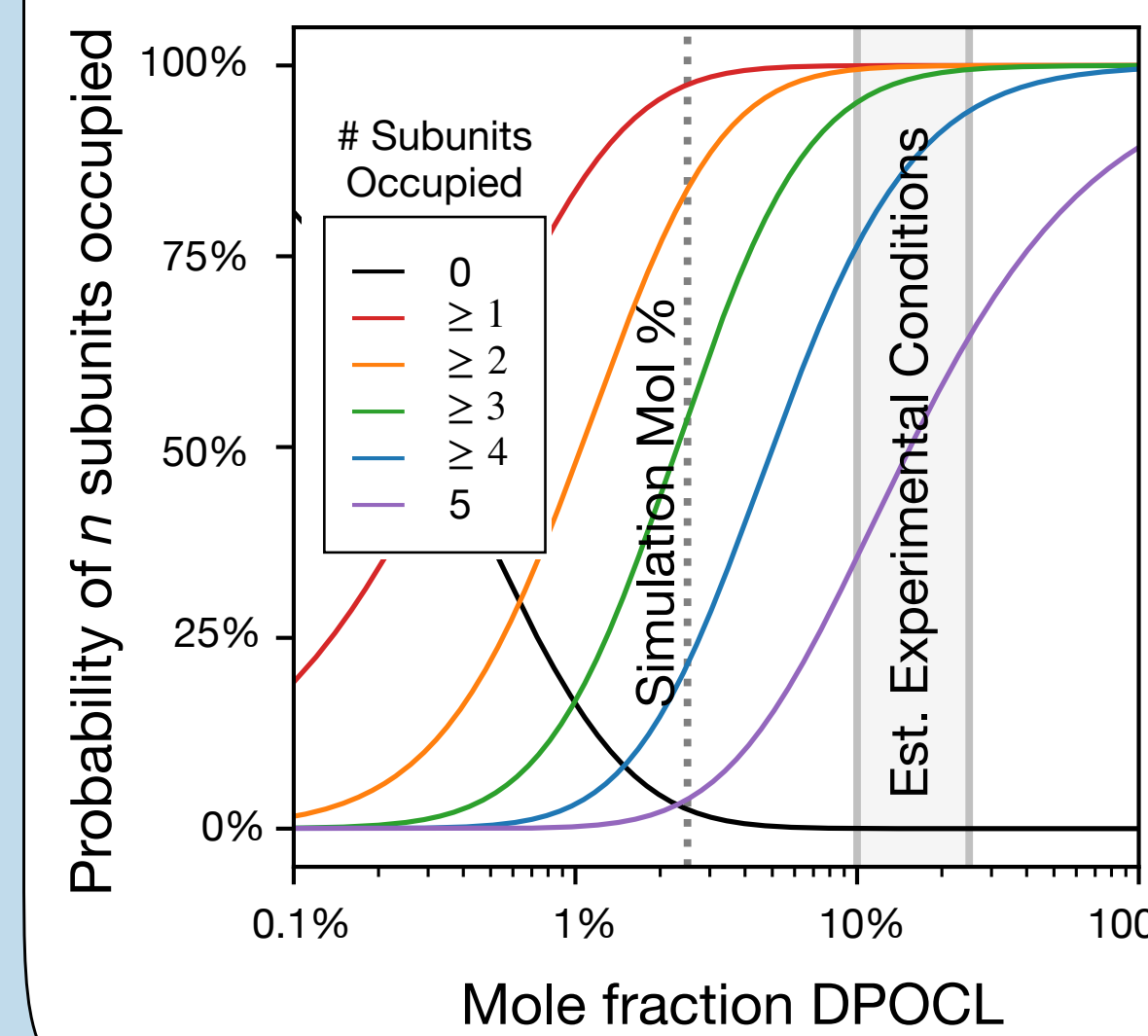


Fig. 2: Estimated probability that at least n intrasubunit sites will be bound simultaneously, assuming binding is an independent event.

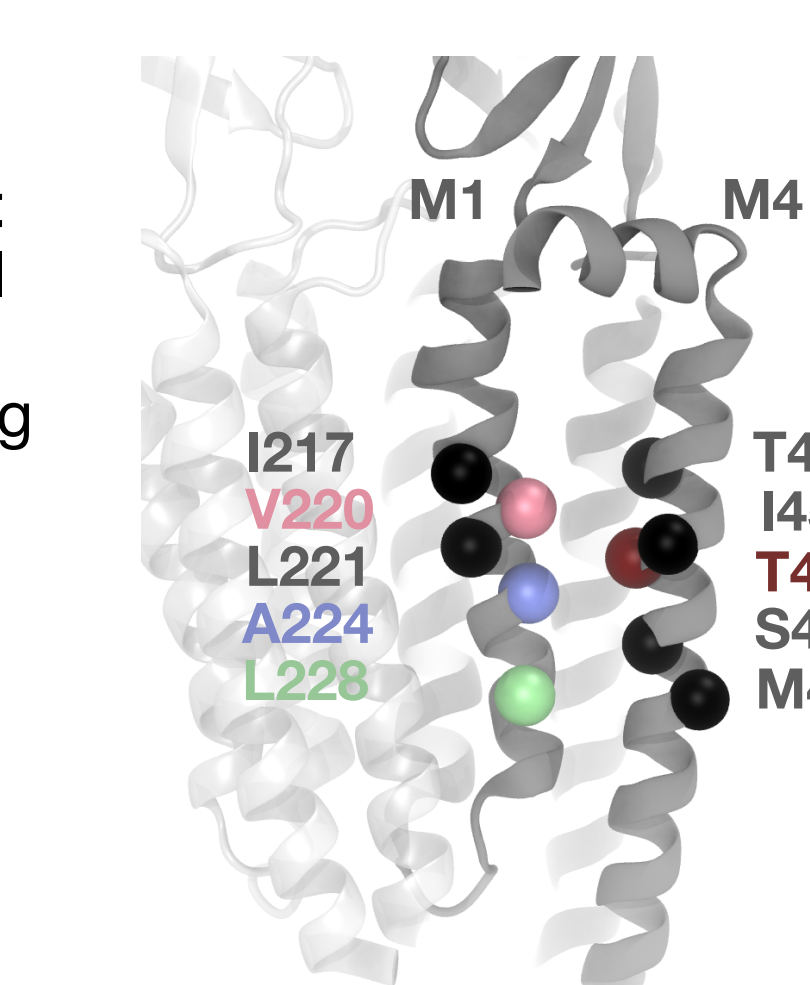


Fig. 4: Amino acid residues in intrasubunit groove that contact polar AA headgroups for greater than 10% of trajectory are highlighted. Each highlighted residue was mutated for further testing.

RESULT 2: Cardiolipin (CDL) preferentially binds the resting state⁷ of the prokaryotic pLGIC ELIC5

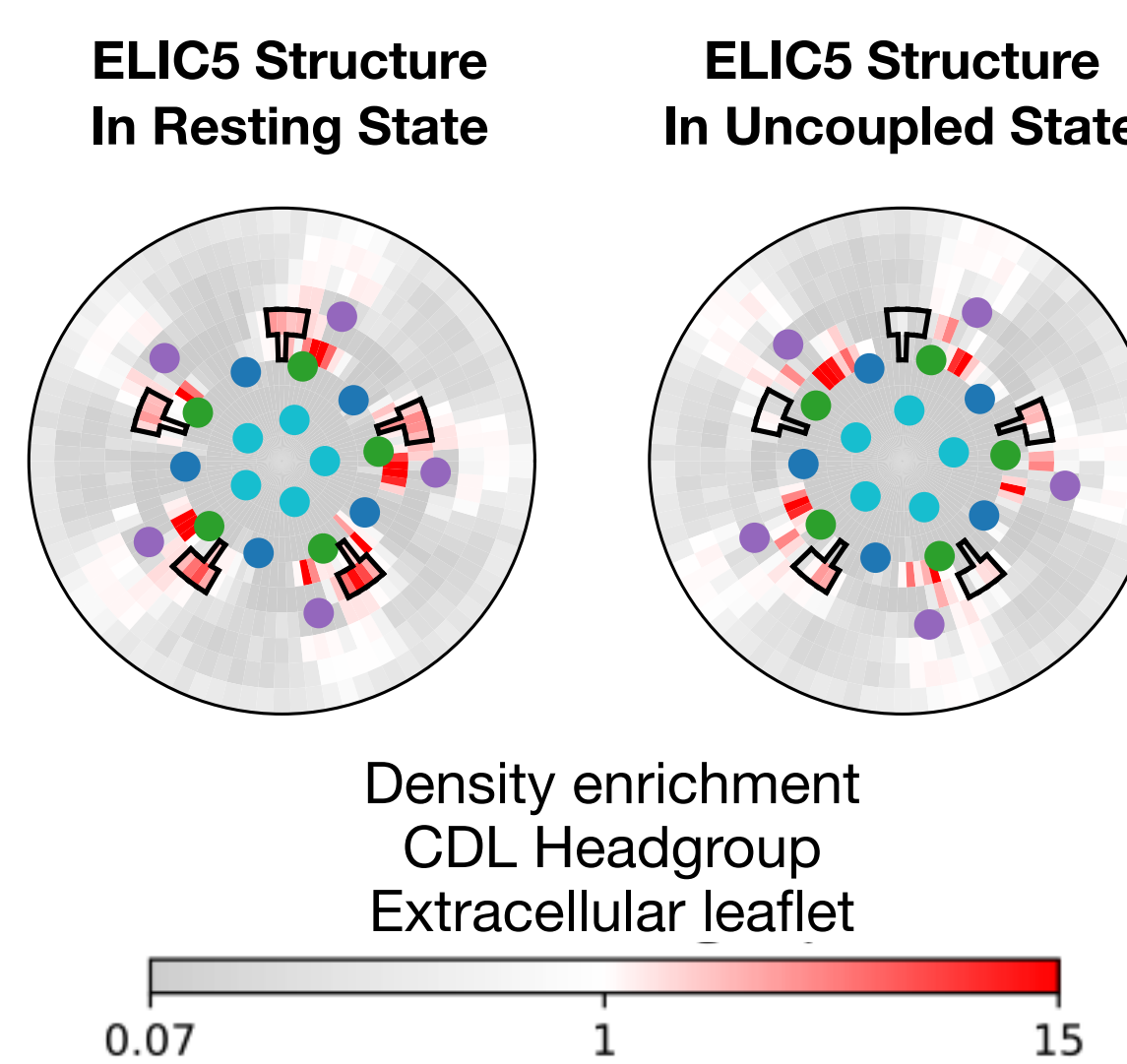
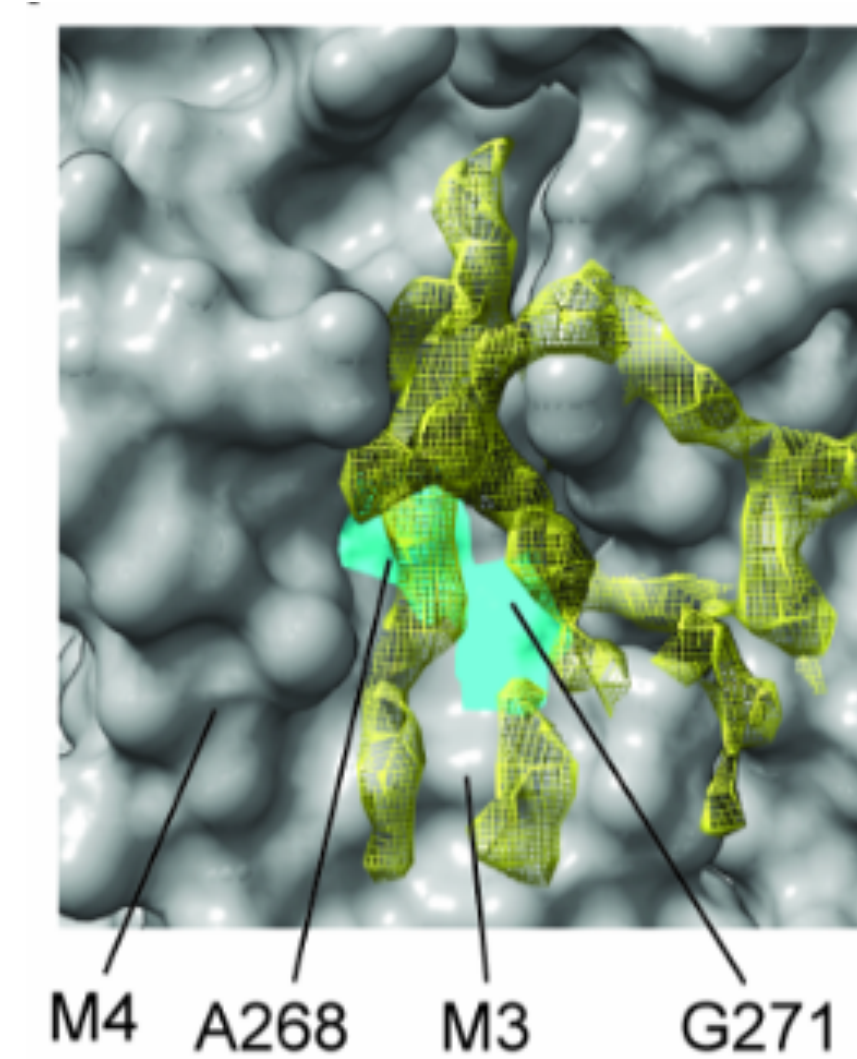


Fig. 6: Four-legged lipid density observed on ELIC5 surface by cryo-em in a 3:1 POPC:CDL nanodisc. This density is presumed to be cardiolipin, occupying the M3/M4 groove in the protein's resting state.

Fig. 7: Density enrichment of CDL headgroups measured in a 20:1 POPC:CDL membrane. Region of high enrichment is outlined in black. Density-Threshold Affinities (ΔG_{bind}) for CDL headgroups in that site are -1.5 ± 0.1 and -0.8 ± 0.1 kcal/mol, in the two respective structures.

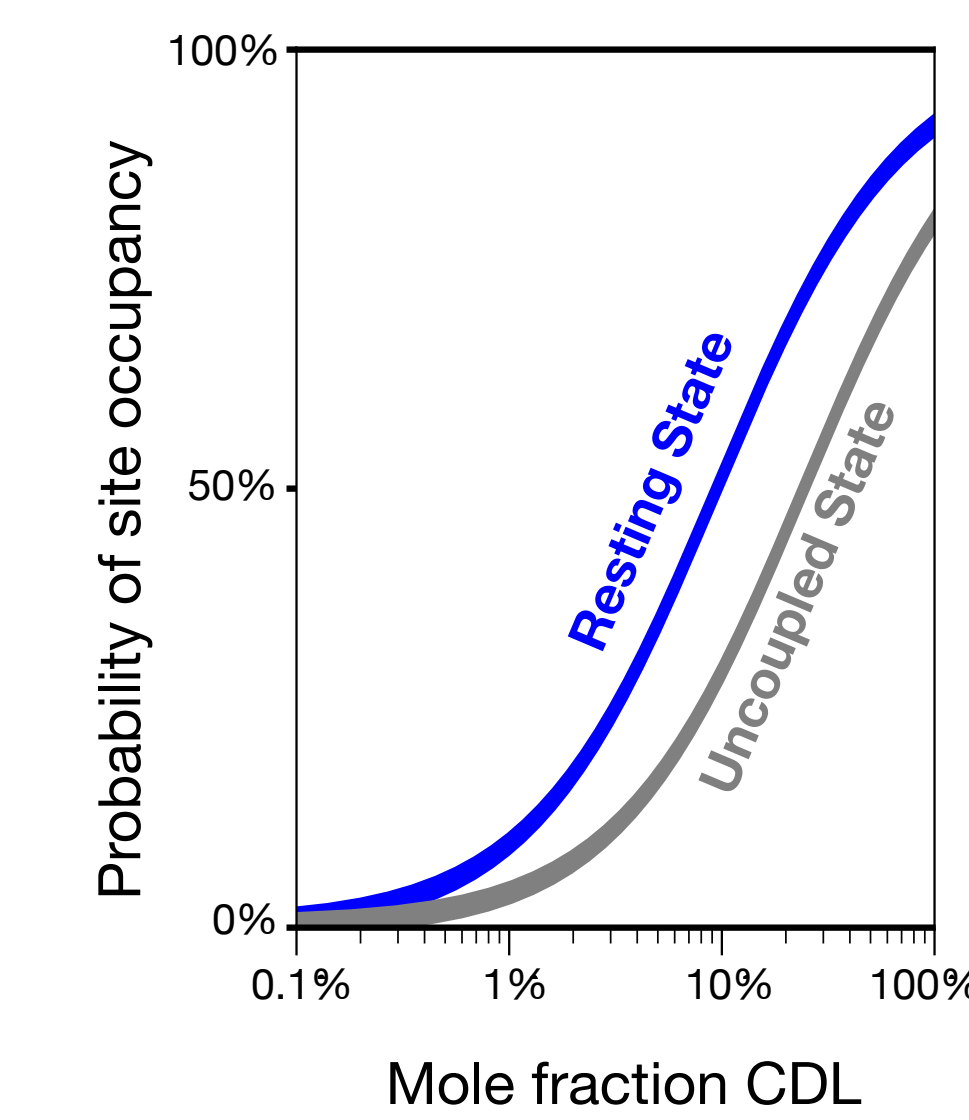


Fig. 8: Estimated probability that at least one of the five intrasubunit sites will be occupied, as a function of mole fraction x_{CDL} .

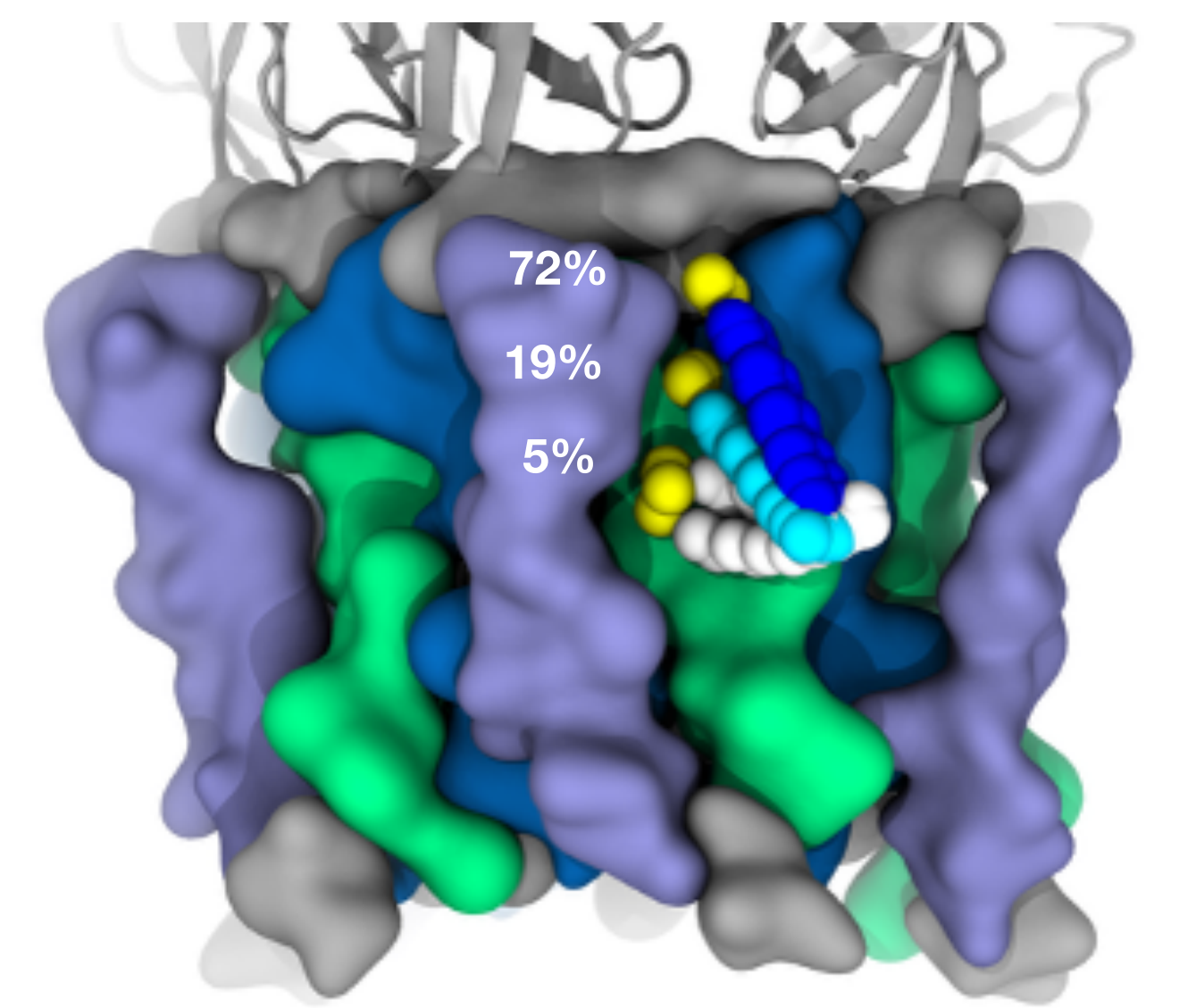
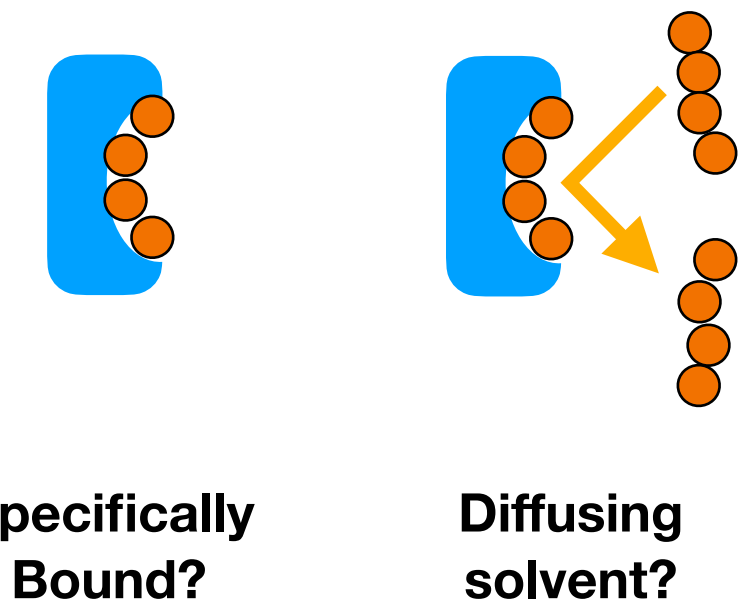


Fig. 9: Most occupied binding modes depict CDL head groups interacting with the M3/M4 intrasubunit groove. The most populated binding mode is consistent with the lipid density observed in Fig. 6.

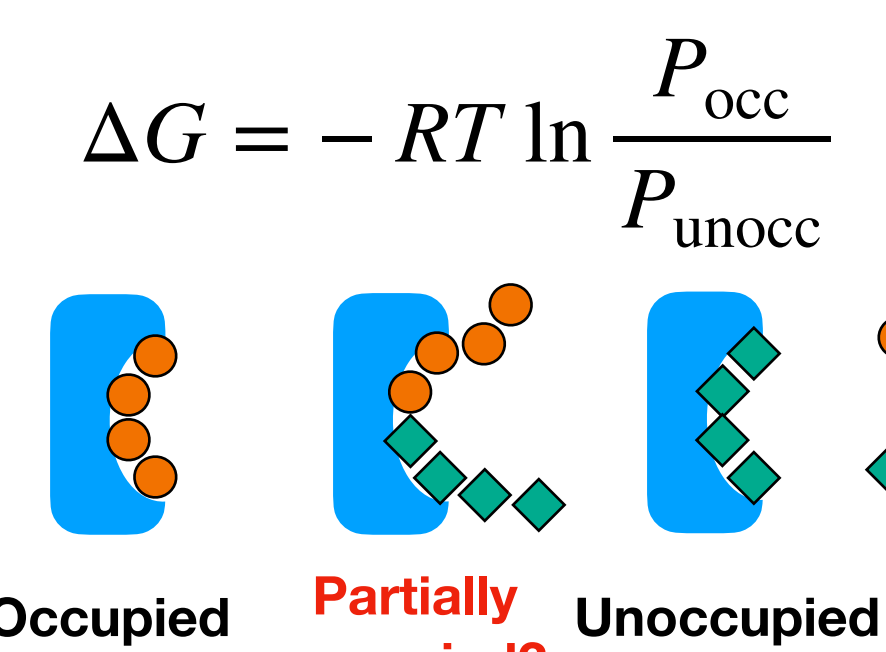
Problems When Defining "Binding" in Membrane Contexts

Problem 1: Ligand and/or Solvent?



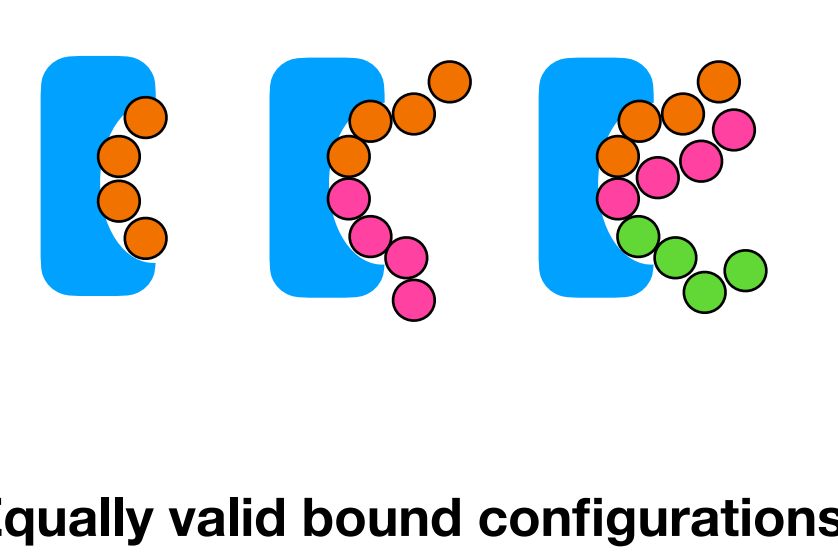
Specifically Bound? Diffusing solvent?

Problem 2: Partial Occupancy



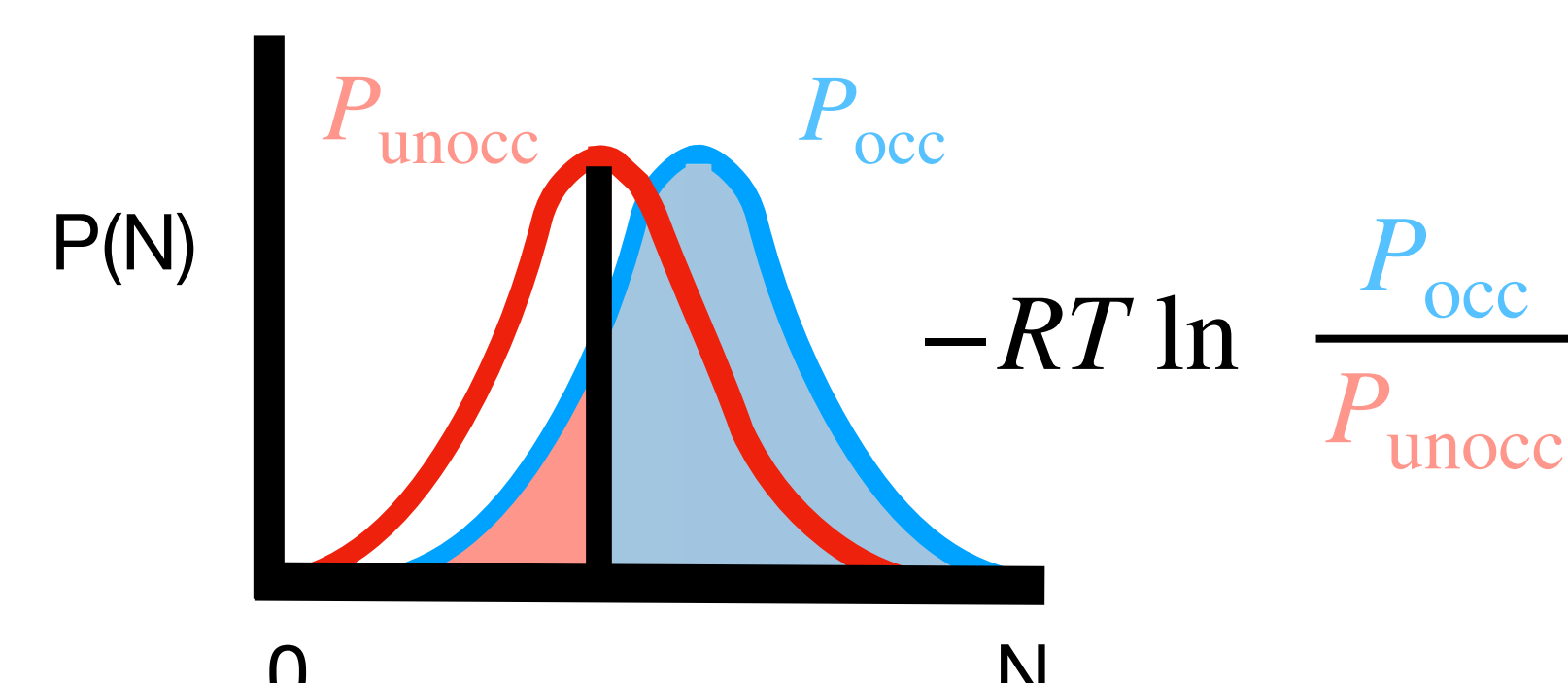
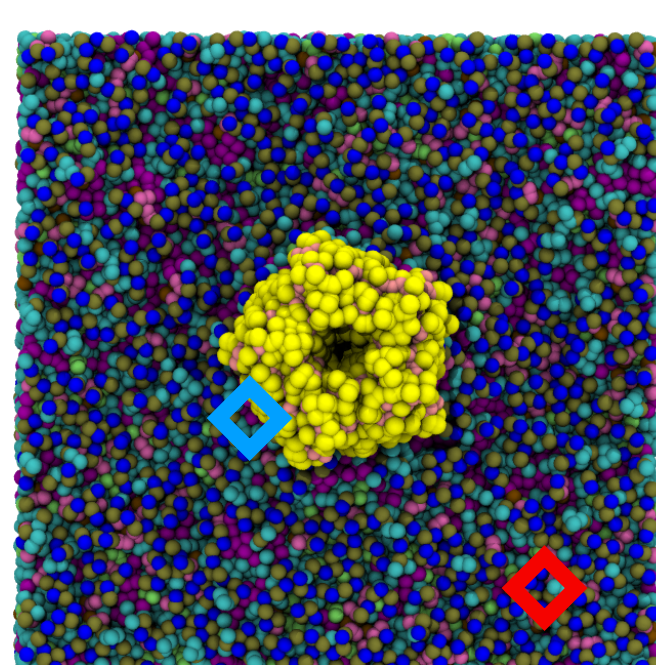
Occupied Partially occupied? Unoccupied

Problem 3: Chemically Indistinct, Chain-like Ligands



Equally valid bound configurations

The Density-Threshold Affinity (DTA)^{1,2}



Use coarse-grained molecular dynamics (CG-MD) simulations to measure the probability of finding an atom of interest in the selected binding site relative to finding it in a bulk patch of the same area. Construct probability distributions for finding N atoms in the site and in the bulk patch. Subdivide the site distribution at the peak of the bulk distribution to get P_{occ} and P_{unocc} .

Methods

MD Software: GROMACS 2024³
Forcefield: MARTINI 2.2⁴

RESULT 1
PDB ID: 8V89⁵
Membrane: Oocyte-mimetic adapted from Hill⁶
Simulations: 4 ind. replicas, 10 μs each

RESULT 2
DB IDs: 9YEN, 9YEW⁷
Membrane: 20:1 POPC:CDL
Simulations: 3 ind. replicas, 25 μs each

$$P_{\text{occ}}(x_{\text{lipid}}) = \frac{x_{\text{lipid}}}{e^{\Delta G_{\text{bind}}/RT} + x_{\text{lipid}}}$$

$$P_{\text{occ},n} = \sum_{i=0}^n \binom{5}{i} (P_{\text{occ}})^i (1 - P_{\text{occ}})^{5-i}$$

Summary

- In both systems, DTA provides molecular insight into experimentally observed effects of lipids on protein structure or function.
- DTA reveals a potential binding site and mechanism of fatty acid inhibition of the nAChR
- DTA shows that cardiolipin may stabilize the resting state of ELIC5, allowing for subsequent activation by agonist molecules.

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Support

NSF DGE 2152059
NSF BIO 2031948
NIH R35GM137957
ERC (Grant no. 788974, Dynacotine)

This work used Delta GPU at NCSA through allocation BIC220103 from the Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support (ACCESS) program, which is supported by U.S. National Science Foundation grants #2138259, #2138286, #2138307, #2137603, and #2138296.